

Project Assignment

ASEN 6062: Celestial Mechanics

Spring 2026

Project Due April 24, 2026.

In this project you will choose a multi-body problem of interest and implement its dynamics and the dynamics of a massless particle driven by it. Thus, in spirit it will be similar to the Computational HW problems 1 and 3, however you will analyze your system in a bit more detail.

Choose a particular restricted problem for the motion of a point mass as attracted by a coherent celestial mechanics system, such as the three body problem, the attraction of a rotating asteroid, the attraction of a binary asteroid, etc. Ideally, you should choose a celestial mechanics system where you can describe its motion analytically, such as the Lagrange solution for the 3-body problem, a synchronously rotating secondary about a primary body for the binary asteroid problem, or following the Hill variation orbit for the motion of the Earth and Moon as perturbed by the sun, for example.

The main part of the project will be describing the motion of a particle as attracted by this system, computing both a nominal trajectory for a range of possible initial conditions, and computing the accompanying State Transition Matrix to describe the motion in the vicinity of the nominal trajectory. The project will then have you test your solution properties, and perform some specific computations of interest.

The specific items that you should follow for this project are listed below. The final product will be a report describing the results:

1. Propose a specific model for study and submit it, with a short description, to Gradescope by March 30 for approval. This is a firm deadline, and approvals should be given by April 2.

The following items are what your final report should contain.

2. For your approved project, state the model for the motion of your system with mass. Explain how you are modeling their motion, and whether they form a relative equilibrium or a time-periodic system.
3. Write the equations of motion for your massless particle as attracted by the bodies with mass. Derive the equations using both Lagrange's

Equations and Hamilton's Equations, and state the transformations between these two sets of coordinates. Also derive the equations of motion for the state transition matrix along your given solution. Be sure to state whether your system is in a rotating frame, inertial frame, etc. Describe which set of equations you will integrate, and the numerical package you plan to use.

4. Perform test integrations to verify your numerical results. In particular, if your system is uniformly rotating you can define the Jacobi integral and test if its constant. For the State Transition Matrix you can test whether it is symplectic when transformed into the Hamiltonian frame, or if in a Lagrangian frame you can verify whether it satisfies the correct eigenvalue properties (discussed in class).
5. Compute some specific solutions of interest for your problem and analyze their stability using the State Transition Matrix. These can be periodic orbits, quasi-periodic orbits, or general trajectories that have an interesting behavior.
6. Discuss the results that you find, and indicate what future analysis would be of interest.

Possible Projects

On Canvas there will be a number of papers and chapters that define possible problems of interest. You can also propose your own problem for your project. Possible ideas include:

- A time-periodic system such as the Elliptic Restricted 3BP and or Hill Restricted 4BP.
- Motion around a non-spherical body, utilizing the ellipsoid or polyhedron potential. The body could be uniformly rotating or following general torque-free rotation (which would create a time-periodic frame in the body-fixed frame).
- Motion about a central configuration which either follows a uniform rotation or a homographic elliptic orbit that creates a time-periodic system.